LIFTING MAPS FROM A PERFECT OBJECT DEFINED ON A QUASI-COMPACT OPEN OF AN AFFINE SCHEME AFTER TENSORING WITH A KOSZUL SWINDLE

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Allowed standard prerequisites (editorial).

- On an affine scheme perfect objects correspond to finite complexes of finite projective modules and derived quasi-coherent objects to module complexes. invoked at proof lines 3589-3593. Provides the module representatives M^bullet and (M')^bullet.
- The internal Hom of the representatives is computed by the Hom complex of the module complexes, and maps on the open U are sections over U of the internal Hom. invoked at proof lines 3594-3600. Turns alpha into an element s of H^0(U, R SheafHom(E, E')).
- Cohomology classes on a quasi-compact open of an affine scheme are represented by maps from the Koszul complexes of powers of the functions cutting out the complement. invoked at proof lines 3601-3606. Provides the map xi from the Koszul complex I^bullet(f_1^e, ..., f_r^e).

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited at proof lines 3590-3591 to represent E by M^bullet; the assembly of E_1 is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 3592-3593 to represent E' by (M')^bullet and again in the final identification; the assembly is inside the counted span.

Source reference. Original lemma 08DM: sources/stacks-chow/tags.txt, source lines 10642–10642. Cited at proof lines 3596-3598 to represent R SheafHom(E, E'); the section argument is inside the counted span.

Source reference. Original lemma 08DK: sources/stacks-chow/tags.txt, source lines 10640–10640. Cited at proof lines 3599-3600 and 3610-3611 for the section s and the assembly of gamma; the construction is inside the counted span.

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Source reference. Original proposition 08DD: sources/stacks-perfect/pinned-original-source.tex, source lines 2316–2316. Cited at proof lines 3603-3606 for the map xi from the Koszul complex; the construction of E_1 is inside the counted span.

Lemma 0.2. *Let X be an affine scheme. Let $U \subset X$ be a quasi-compact open. Let E, E' be objects of $D_{QCoh}(\mathcal{O}_X)$ with E perfect. For every map $\alpha : E|_U \to E'|_U$ there exist maps*

$$E \xleftarrow{\beta} E_1 \xrightarrow{\gamma} E'$$

of complexes on X with E_1 perfect such that $\beta : E_1 \to E$ restricts to an isomorphism on U and such that $\alpha = \gamma|_U \circ \beta|_U^{-1}$. Moreover we can assume $E_1 = E \otimes^{\mathbf{L}}_{\mathcal{O}_X} I$ for some perfect complex I on X.

Proof. Write $X = \mathrm{Spec}(A)$. Write $U = D(f_1) \cup \ldots \cup D(f_r)$. Choose finite complex of finite projective A-modules $M^\bullet$ representing E (Lemma 08EB). Choose a complex of A-modules $(M')^\bullet$ representing E' (Lemma 06Z0). In this case the complex $H^\bullet = \mathrm{Hom}_A(M^\bullet, (M')^\bullet)$ is a complex of A-modules whose associated complex of quasi-coherent $\mathcal{O}_X$-modules represents $R\mathcal{H}om(E, E')$, see Cohomology, Lemma 08DM. Then α determines an element s of $H^0(U, R\mathcal{H}om(E, E'))$, see Cohomology, Lemma 08DK. There exists an e and a map

$$\xi : I^\bullet(f_1^e, \ldots, f_r^e) \to \mathrm{Hom}_A(M^\bullet, (M')^\bullet)$$

corresponding to s, see Proposition 08DD. Letting E_1 be the object corresponding to complex of quasi-coherent $\mathcal{O}_X$-modules associated to

$$\mathrm{Tot}(I^\bullet(f_1^e, \ldots, f_r^e) \otimes_A M^\bullet)$$

we obtain $E_1 \to E$ using the canonical map $I^\bullet(f_1^e, \ldots, f_r^e) \to A$ and $E_1 \to E'$ using ξ and Cohomology, Lemma 08DK. □